

Reproducible Data Provenance for an Investigational New Drug Submission

Drug candidate: TSR-X2 (CAR-T responder prediction model, demonstration)

This document proves the exact data provenance of the machine-learning model that designed the drug candidate named above. It is generated automatically from the Tessera Bio content-addressable release bundle identified by the release hash below. Every artefact in the bundle is cryptographically pinned; the bundle can be reproduced byte-for-byte by any reviewer using the *tessera checkout* command on the final page of this report.

SPONSOR	Tessera Bio / ReadyOmics (demonstration)
TARGET IND DATE	2027-04-15
RELEASE BUNDLE	tessera.multiomic.v0.2.2026-09.cptac-breast.ensembl-v110
GENERATED	2026-09-18 07:57 UTC

release_id: tessera-release:f6bf1a549aca

1. Executive Summary

This audit trail report establishes the cryptographic chain of custody between the training data used by the machine-learning model that designed **TSR-X2 (CAR-T responder prediction model, demonstration)** and the model's output. It is generated automatically by the Tessera Bio platform from a single content-addressable release bundle, identified by the release hash recorded in Section 7 of this document. The report is intended to accompany the Investigational New Drug (IND) application submitted by Tessera Bio / ReadyOmics (demonstration) on 2027-04-15.

The Tessera Bio platform implements the requirements of 21 CFR Part 11 (Electronic Records; Electronic Signatures) by treating every dataset release as a content-addressable, cryptographically-signed artefact. The release hash, computed as the SHA-256 of a canonical JSON manifest of all bundled artefacts (matrices, splits, identifier-drift report, provenance manifest), provides a tamper-evident binding between the training data and any downstream model output. Any modification of any artefact — even a single byte — produces a different release hash and is immediately detectable at verification time.

This report is generated on 2026-09-18 07:57 UTC from release bundle `tessera-release:f6bf1a549aca`. The release hash (long form) is recorded in Section 7. The verification procedure any reviewer can use to reproduce this exact byte-for-byte bundle is recorded in Section 8.

2. Release Identification

The release bundle is the unit of audit. It is identified by a single release hash that pins together all artefacts required to reproduce the model's training data. The bundle's identity is recorded below.

Field	Value
Dataset ID	tessera.scrna.v0.3.2026-09.cart-responder.ensembl-v110
Dataset version	0.3.0
Release date	2026-09-18
Producer	Tessera Bio / ReadyOmics
Licence	CC-BY 4.0 (Tessera aggregate; see source_attribution)
Schema version	0.3.0
Release ID (short)	tessera-release:f6bf1a549aca
Release hash (long)	tessera-release:f6bf1a549acaba19d21b9cfeae04a58ce728b06fbe0230f50ee3b5249

The release hash is computed as the SHA-256 of the canonical JSON manifest of all artefacts in the bundle. The manifest is sorted alphabetically by artefact name and serialised with no whitespace, producing a deterministic, order-independent hash. The short form (`release_id`) is the first 12 hex characters of the long form, analogous to a git commit short hash.

3. Data Provenance Chain

The model that designed the drug candidate was trained on a Tessera Bio ML-ready dataset. The dataset is constructed from publicly available primary sources, harmonised against pinned reference databases, split into leakage-controlled train/validation/test partitions, and released as a content-addressable bundle. Each step in the chain is recorded below.

3.1 Cohort definition

Field	Value
Species	Homo sapiens
Tissue	peripheral blood mononuclear cells (PBMC)
Disease focus	B-cell acute lymphoblastic leukemia (CAR-T therapy)
Total samples	24
Total donors	24
Selection criteria	Pediatric and young adult B-ALL patients receiving CD19 CAR-T (tisagenlecle...

3.2 Input sources and licence attribution

Every primary source contributing samples to this dataset is recorded with its original licence and submitter acknowledgement. Tessera Bio preserves source-level attribution through the entire pipeline so that downstream users can verify the licence terms under which each sample was contributed.

Source ID	URL	Licence	N samples	Submitter acknowledgement
GEO-GSE240XXX-CART-RESPONDER	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE240XXX	Public domain (US GDS 240XXX)	24	Submitter, 2024 Nature Medicine

3.3 Reference version pinning

The dataset is pinned to specific Ensembl and UniProt releases. The pins are immutable for the lifetime of this release bundle. New reference releases produce a new dataset card with a new release hash — they do not silently overwrite this one. This is the mechanism by which Tessera Bio provides cross-release identifier stability, the second of the three platform differentiators (U2).

Reference	Pin
Ensembl release	v110 (GRCh38)
UniProt release	202401
Gene ID type	Ensembl Gene ID (v110)
Transcript ID type	Ensembl Transcript ID (v110)

4. Identifier Drift Status

Tessera Bio computes an identifier-drift report on every release. The report quantifies the changes between the Ensembl release used for this dataset and the immediately preceding release. The report is part of the release bundle and is itself hash-pinned — modifying it would change the release hash and fail verification.

Drift metric	Value	Interpretation
Compared to Ensembl release	109	The prior release against which this release is compared
Genes added	3017	New identifiers in the current release not present in the prior
Genes removed	0	Identifiers present in the prior release, retired in the current r
Genes merged	0	Distinct identifiers in the prior release that are now a single id
Genes split	0	A single identifier in the prior release that is now multiple ide
Impact on this dataset	v0.3 pinned to Ensembl v110 + CellTypist v1.5; no impact on this release	Ensembl v110 + CellTypist v1.5; no impact on this release

For this release, the drift impact is recorded as: v0.3 pinned to Ensembl v110 + CellTypist v1.5; no impact on this release. Cell type labels are stable under the pinned taxonomy.. The model that designed the drug candidate was trained against the pinned identifiers in this release. Any future analysis claiming to reproduce this model's behaviour must use the same pinned identifiers, not the current Ensembl release. The tessera checkout command in Section 8 reproduces the exact identifier state at training time.

5. Leakage Controls

A common failure mode in biological machine learning is donor overlap between training and test partitions: a model appears to generalise well, but the test set contains samples from donors also present in training. Tessera Bio enforces biologically-stratified, leakage-controlled splits by default. The split strategy and the verification results are recorded below and are part of the hash-pinned release bundle.

5.1 Split strategy and partition sizes

Partition	N samples	Cohort IDs
Train	14	CART-RESPONDER-TRAIN
Validation	4	CART-RESPONDER-VAL
Test	6	CART-RESPONDER-TEST

Split strategy: held_out_donors_cells (all cells from a donor go to one split; no donor crosses train and test)

5.2 Leakage verification

Check	Result
Donor overlap check	passed (zero donor IDs shared across splits)
Family overlap check	passed (no two donors from the same family in different splits)

6. Batch Effects

Batch effects — systematic differences between samples processed in different wet-lab runs — are a major source of irreproducibility in biological machine learning. Tessera Bio quantifies batch effects before and after correction, and the correction method is recorded as part of the hash-pinned release bundle.

Field	Value
Batch effect detected	True
Correction method	harmony 0.0.9 across 10x chemistry batches + library-prep batch
PCA variance explained by batch (before)	0.43
PCA variance explained by batch (after)	0.05

7. Cryptographic Signatures and Release Hash

The release bundle is signed. The signature, the dataset card hash, and the output matrix hash are all recorded below. In production these signatures are produced via Sigstore / cosign; in this demonstration release they are produced via the Tessera Bio deterministic SHA-256 signing stub, which is a placeholder for the production signing key.

Field	Value
Output matrix SHA-256	sha256:5c4e8a9f0b1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e
Dataset card SHA-256	sha256:789abc012def345ghi678jkl90lmno234pqr567stu890
Signed by	tessera-release-signing-key-2026
Release ID (short, 12 hex chars)	tessera-release:f6bf1a549aca
Release hash (long, 64 hex chars)	tessera-release:f6bf1a549acaba19d21b9cfeae04a58ce728b06f6be0230f50ee3b

7.1 Bundled artefacts (full manifest)

The release hash is computed over the following artefacts. Each artefact's individual SHA-256 is recorded; the release hash is the SHA-256 of the canonical JSON of this list. Verification re-hashes every artefact and recomputes the release hash from scratch.

Artefact path	SHA-256 (first 24 hex chars)	Size (bytes)
dataset_card.yaml	b12212150300cec23016203c...	8626
identifier_drift_report.json	d18ab3acf7fa3c5ad5953024...	127
modalities/single-cell-rna-seq.h5ad	29d285f5d691baff9b1efdb1...	294
provenance_manifest.json	a550ad0cce460020d8f46644...	48
splits/test.parquet	7da36c5853a7f62709a540a0...	83
splits/train.parquet	fdae423c1419fe367519ff46...	84
splits/validation.parquet	65dcc4f742582873bb343e89...	82

8. Reproducibility Statement

Any reviewer can reproduce the exact byte-for-byte release bundle used to generate this audit trail using the Tessera Bio command-line interface. The checkout command below verifies every artefact's SHA-256, recomputes the release hash, and fails closed if any byte differs from the values recorded in Section 7.

8.1 Verification procedure

```
$ tessera checkout \
  --source-dir ./cptac-breast-multimomic \
  --dest-dir ./audit-verification \
  --expected-hash tessera-release:f6bf1a549aca

OK – bundle checked out at ./audit-verification
  release_id   : tessera-release:f6bf1a549aca
  release_hash : tessera-release:f6bf1a549acaba19d21b9cfeae04a58ce728b06fbe0230f50ee3b5249
f875c68
  artefacts    : 7 (all hashes verified)
```

8.2 Tamper-evidence guarantee

If any artefact in the source directory has been modified after the release was created, the checkout command will fail with a precise artefact-level error naming the file whose hash differs from the recorded value. This behaviour is verified by the Tessera Bio unit test suite (test_release.py, 11 tests including tamper-detection and byte-identical-checkout tests) which is run on every change to the platform codebase.

8.3 Audit-relevant Part 11 controls addressed

Part 11 section	Requirement	How Tessera addresses it
11.10(a)	Validation of systems	Tessera's release/checkout pipeline is validated by 11 automated unit tests
11.10(c)	Record retention and protection	Every release bundle is a self-contained directory whose integrity is verified
11.10(d)	Limit system access	Production deployments use Sigstore / cosign for signing keys; ABAC for access control
11.50	Computer-generated, time-stamped audit trails	Every release carries a release_date, signing_key identifier, and per-artefact hashes
11.70	Signature / record linking	The release hash cryptographically binds the signature to the signed release

9. Sign-off

By signing below, the named reviewer confirms that they have inspected the release bundle identified in Section 7 of this report, that the release hash recorded in Section 7 matches the release hash recomputed from the bundle in their possession, and that the bundle constitutes the complete and unmodified training data used by the model that designed TSR-X2 (CAR-T responder prediction model, demonstration).

9.1 Prepared by (data engineering)

Name	_____	Date	_____
Role	Data Engineering Lead	Signature	_____

9.2 Reviewed by (QA)

Name	_____	Date	_____
Role	Quality Assurance Reviewer	Signature	_____

9.3 Approved by (Regulatory Affairs)

Name	_____	Date	_____
Role	Regulatory Affairs Signatory	Signature	_____

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